

|| Jai Sri Gurudev ||
Sri Adichunchanagiri Shikshana Trust®

SJC INSTITUTE OF TECHNOLOGY

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P.B. No. 20, B.B. Road, Chikkaballapur - 562 101, Karnataka.

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Estd. 1986

Department of Electronics and communication engineering

ACTIVITY BASED LEARNING ASSIGNMENT

TECHNICAL SEMINAR REPORT

TOPIC: FM RADIO RECEIVERS

SUBMITTED BY

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BRANCH : ECE-C

UNDER THE GUIDANCE OF

MENTOR NAME:Dr.HAREESH K KUMAR

DESIGNATION: ASSOCIATE PROFESSOR

DEPARTMENT: ECE

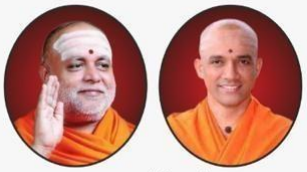


DEPARTMENT OF FIRST YEAR ENGINEERING

ACADEMIC YEAR 2025-26

EVEN SEMESTER

MAY-2026



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CERTIFICATE

This is to certify that Mr /Ms. **VACHANA G,CHATHUSH,SHREYAS,VIKAS ADITYA GOWDA P** Bearing USN:**1SJ25EC153, 1SJ25EC152, 1SJ25EC134, 1SJ25EC162** of **ELECTRONICS AND COMMUNICATION ENGINEERING** branch **2nd** semester has successfully submitted activity-based assignment Technical Seminar Report on the topic: **FM RADIO RECEIVERS** For the year Academic year 2025-26.

Subject teacher **Cycle Co-ordinator** **Co-ordinator,FYE**

ABA REPORT
ON

FM RADIO RECEIVERS

Department: Electronics and Communication Engineering

Submitted By
SHREYAS

Semester: 2nd SEMESTER **Class and Section: ECE-C**

Marks:

SI NO	USN	Name	1	2	3	4	Total (10)
1.	1SJ25EC134	SHREYAS					

Sl. No	Rubrics	Marks
1.	Understanding of Concept and Theory	3
2.	Program Implementation and Correctness	3
3.	Output Results, Graphs and Analysis	2
4.	Presentation and Viva Voce	2
Total		**10 Marks**

Faculty,
Dr HAREESH K KUMAR,
ASSOCIATE PROFESSOR,
DEPARTMENT OF ECE,
SJC INSTITUTE OF TECHNOLOGY,
CHICKBALLAPUR-562 101

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INTRODUCTION:

FM (Frequency Modulation) Radio Receiver is an electronic communication system designed to receive radio signals transmitted using frequency modulation and convert them into clear audio output. In FM transmission, the frequency of the carrier wave changes according to the amplitude of the input audio signal, while the amplitude of the carrier remains constant. This method reduces the effect of noise and provides high-quality sound reception.

The FM radio receiver captures transmitted radio waves through an antenna and processes them through various stages to reproduce the original audio signal. The antenna receives electromagnetic waves and sends them to the tuner section, which selects the desired radio station frequency. The selected signal is then amplified to increase its strength. The demodulator extracts the original audio information from the FM signal, and the audio amplifier further increases the signal power before sending it to the speaker for sound output.

METHODOLOGIES:

The methodology of an FM radio receiver describes the sequence of operations used to receive and convert radio signals into audio output.

Signal Reception

The antenna receives Frequency Modulated (FM) radio waves transmitted from broadcasting stations.

RF Amplification

The received weak radio frequency signal is amplified by the RF amplifier to improve signal strength and sensitivity.

Tuning Process

The tuner selects the desired FM station frequency and rejects unwanted frequencies.

Frequency Conversion (Mixer Stage)

The selected signal is mixed with a local oscillator signal and converted into an Intermediate Frequency (IF), usually 10.7 MHz.

IF Amplification

The intermediate frequency signal is amplified to increase signal strength and improve selectivity.

FM Detection / Demodulation

The FM detector extracts the original audio information from the modulated FM signal.

Audio Amplification

The extracted audio signal is amplified using an audio amplifier.

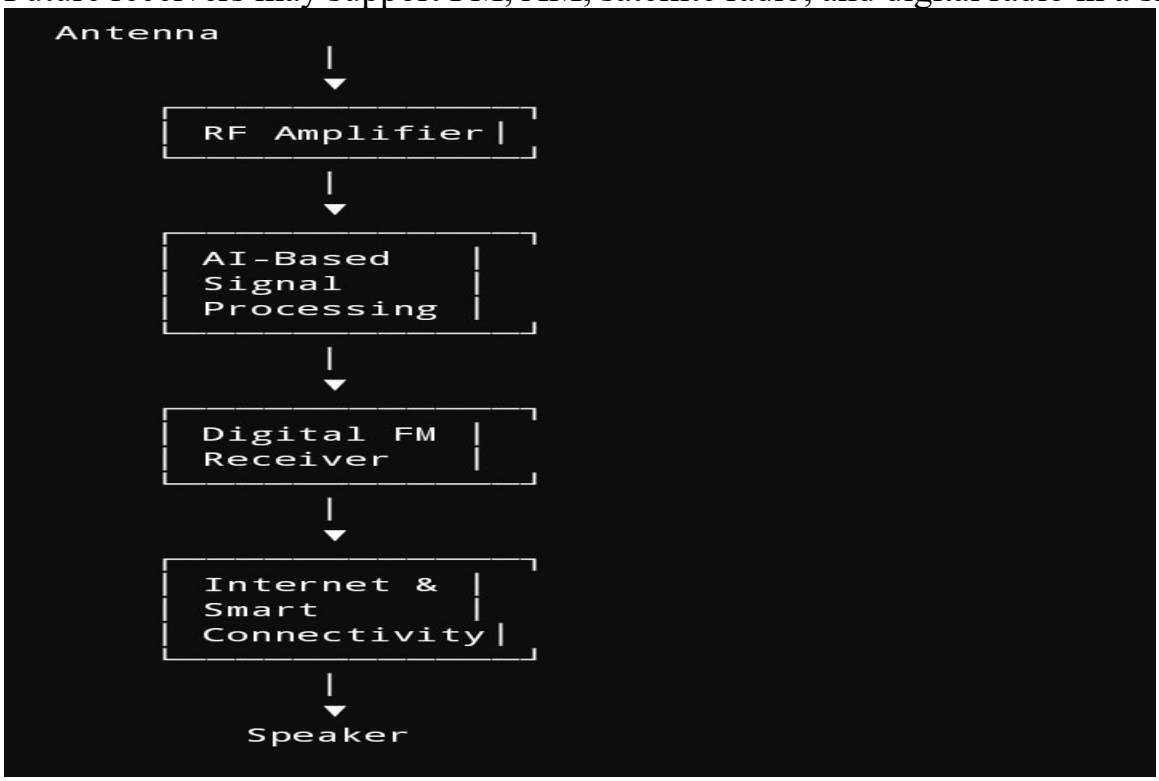
Sound Output

The amplified signal is sent to a speaker or headphones to produce sound.

FUTURE SCOPE:

FM radio receivers continue to evolve with modern communication technologies. Future improvements can make them smarter, more efficient, and more versatile.

Future Scope:
Digital FM Technology
Future FM receivers may use digital broadcasting technology for better sound quality and reduced noise.
AI-Based Signal Processing
Artificial Intelligence can automatically tune frequencies and improve signal reception.
Internet Integration
FM receivers can be integrated with internet radio for access to worldwide stations.
Mobile and Smart Device Connectivity
Future systems may connect directly with smartphones and smart home devices.
Low Power Consumption
Advanced designs can reduce battery usage and increase efficiency.
Miniaturization
Compact chips can make FM receivers smaller and portable.
Multi-band Reception
Future receivers may support FM, AM, satellite radio, and digital radio in a single device.



ADVANTAGES AND DISADVANTAGES:

Advantages:

Better sound quality with high fidelity audio.

Less affected by noise and signal interference.

Provides clear and distortion-free reception.

Suitable for music broadcasting due to good audio quality.

Large coverage area for local broadcasting stations.

More stable signal reception compared to AM in noisy environments.

Disadvantages:

Requires a larger bandwidth compared to AM systems.

Limited transmission range because FM signals travel mostly in line-of-sight.

Reception quality can decrease due to obstacles like buildings or mountains.

FM receiver circuits are more complex than AM receivers.

Equipment cost can be slightly higher.

Weak signals may cause fading and poor reception**.**

CONCLUSION:

Smartphone charger design demonstrates the practical application of rectifiers, filters, and regulators in power electronics. By converting AC mains into smooth 5V DC, chargers ensure safe and efficient charging. With advancements like **GaN technology, fast charging, and wireless systems**, the future of charger design promises even greater efficiency, compactness, and intelligence.

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7.GEO TAG PHOTO:

